Amir Beck, Chapter 1 - Exercise 1.6

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14 October 2026

1.6. Let $||.||$ be a norm on $\mathbb{R}^n$. Show that the norm function $\mathbf{f}(\mathbf{x}) = ||\mathbf{x}||$ is a continuous function over $\mathbb{R}^n$

We have to show that, for all $\mathbf{a} \in \mathbb{R}^n$,

$$\forall \varepsilon > 0 \exists \delta > 0 ||\mathbf{x} - \mathbf{a}||_{\mathbb{R}^n} < \delta \implies ||f(\mathbf{x}) - f(\mathbf{a})||_{\mathbb{R}} < \varepsilon,$$

that is,

$$\forall \varepsilon > 0 \exists \delta > 0 ||\mathbf{x} - \mathbf{a}||_{\mathbb{R}^n} < \delta \implies \Big| ||\mathbf{x}|| - ||\mathbf{a}|| \Big| < \varepsilon,$$

Let $\mathbf{a} \in \mathbb{R}^n$. Let $\varepsilon > 0$. Take $\delta = \varepsilon$. Thus, whenever $||\mathbf{x} - \mathbf{a}|| < \delta$, we have that

$$\Big| \; ||\mathbf{x}|| - ||\mathbf{a}|| \; \Big| \leq ||\mathbf{x} - \mathbf{a}|| < \delta = \varepsilon$$

Therefore, the norm function is continuous over $\mathbb{R}^n$.

In the above argument we used the fact that $\Big| \; ||\mathbf{x}|| - ||\mathbf{a}|| \; \Big| \leq ||\mathbf{x} - \mathbf{a}||$. In fact, one has

$$||\mathbf{x}|| = ||(\mathbf{x} - \mathbf{a}) + \mathbf{a}|| \leq ||\mathbf{x} - \mathbf{a}|| + ||\mathbf{a}|| \Leftrightarrow \boxed{||\mathbf{x} - \mathbf{a}|| \geq ||\mathbf{x}|| - ||\mathbf{a}||}$$

and

$$||\mathbf{a}|| = ||(\mathbf{a} - \mathbf{x}) + \mathbf{x}|| \leq ||\mathbf{a} - \mathbf{x}|| + ||\mathbf{x}|| \Leftrightarrow \boxed{-||\mathbf{x} - \mathbf{a}|| \leq ||\mathbf{x}|| - ||\mathbf{a}||}.$$